

Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues

Shuhao Liu^{a,†}, Zhuo Chen^{b,†,*}, Zhi Ling^c, Yu Yan^c, Jie Liu^a, Qiuxue Wu^a,
Ziyi Kuang^c

^a*Harbin University of Science and Technology, No. 52 Xuefu Road, Nangang District,
Harbin 150080, Heilongjiang, China*

^b*Chalmers University of Technology, SE-412 96 Gothenburg, Sweden*

^c*Kiwiar Co., Ltd., Singapore*

Abstract

Fixed-capacity review is useful only when distinct plants remain represented after candidate formation, role qualification, ranking, and truncation. We present an evaluation contract for finite crop–weed review queues that separates these losses. At a fixed queue capacity and matching threshold tuple, one-to-one support measures the maximum number of distinct candidate–plant records, whereas set coverage measures the union of truth instances incident to qualified candidates. These are threshold-conditioned geometric endpoints, not bounds on human recovery. On the official WeedsGalore split, proposal operating-point commissioning increased raw spatial support but produced ranker- and contract-dependent queued effects. Under the commissioned-only ranker, one-to-one support increased by 0.0187 at matched burden. Mask R-CNN had the highest point estimate at $K = 20$ (0.2159), although crossed seed-by-image intervals for its contrast with U-Net included zero. Standard AP, AR, and a queue-level PQ-style diagnostic did not expose the 105-instance merge-capacity gap. Two independent operators showed high category agreement, while human reviewability and mask qualification remained non-equivalent. A locked transfer to 60 CWFID images produced no

*Corresponding author.

Email addresses: 2420610137@stu.hrbust.edu.cn (Shuhao Liu),
zhuoc@chalmers.se (Zhuo Chen), zhi.ling@kiwiar.com (Zhi Ling),
yu.yan@kiwiar.com (Yu Yan), liujie@hrbust.edu.cn (Jie Liu),
2420610140@stu.hrbust.edu.cn (Qiuxue Wu), ziyi.kuang@kiwiar.com (Ziyi Kuang)

[†]These authors contributed equally to this work.

candidates from five U-Net checkpoints; post-lock continuous scores showed weak, checkpoint-variable ranking and poor role qualification at low thresholds. The contract therefore supports executable offline quality assurance while making model, matching, and transfer boundaries explicit.

Keywords: crop–weed segmentation, instance evaluation, review queue, matching contract, quality assurance, precision agriculture

1. Introduction

Site-specific weed management links sensing, plant localisation, crop–weed discrimination, and intervention [1, 2]. Field-scale mapping protocols must align imaging, ground validation, and the management question [3]. The same alignment is needed when predicted regions enter an offline review queue. A high pixel score does not establish how many distinct plants remain available when only a finite number of regions can be inspected.

Standard instance metrics answer complementary questions. COCO-style average precision and average recall evaluate detection and mask quality across intersection-over-union thresholds and detection limits [4]. Panoptic Quality combines recognition and segmentation quality after one-to-one matching [5]. Recent analyses show that average precision can obscure duplicate predictions and can change abruptly with matching thresholds [6, 7]. These metrics are essential for model comparison, but they do not encode an agricultural review budget or the sequence by which a plant disappears from a queue.

Finite review introduces several loss mechanisms. A plant can be absent from the proposal pool, contained in a mixed crop–weed region, merged with neighbours, ranked below the cut-off, or duplicated by several candidates. Object-proposal research relates recall to overlap and proposal count [8]. Precision-at-the-top methods address ranking under restricted capacity [9]. An operational audit still needs a contract that connects proposal support, role evidence, multiplicity, ranking, and the number of records available for inspection.

The queue studied here supports offline annotation quality assurance and model-failure review. An operator can retain a region, reject it, or escalate an ambiguous mixture. The queue does not issue an actuation command. This task differs from active learning, which selects observations for annotation or retraining [10, 11]; selective prediction, which abstains from automated

outputs [12, 13]; and learning to defer, which assigns cases between a model and an expert [14, 15].

We present a bounded instance-support evaluation for finite crop–weed review queues. It has four contributions. First, it separates spatial, role-qualified, set-coverage, and one-to-one endpoints. Second, it tests proposal effects across 27 matching contracts and four ranker-training schemes. Third, it combines seed and test-image variation and compares the queue endpoint with standard AP, AR, and PQ-style diagnostics. Fourth, it records human category agreement and a locked cross-dataset transfer. The contribution is an agricultural evaluation contract and evidence decomposition, not a new segmentation architecture or an operational control system.

2. Materials and methods

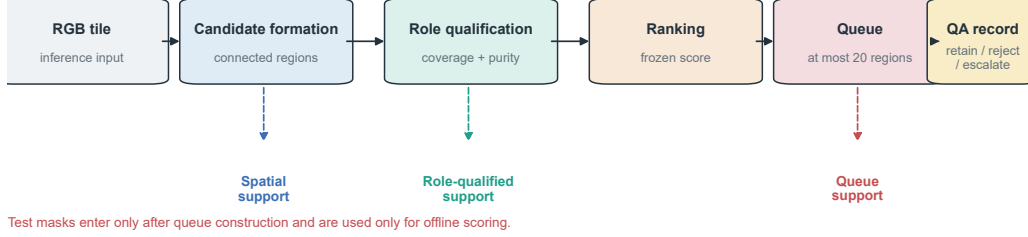
2.1. Review task and endpoint bounds

Each review item contained one RGB tile, one highlighted connected region, and a local RGB view. The intended output was one quality-assurance record: retain, reject, or escalate. Each tile contributed at most $K = 20$ regions. This value standardised capacity across systems. It was not derived from operator time, labour cost, or hectares covered.

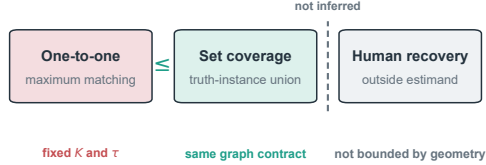
One-to-one support reflects the record contract. For a fixed queue capacity K and matching threshold tuple, a selected region can contribute at most one edge to the maximum-cardinality candidate–instance matching. Set coverage is the number of truth instances incident to at least one edge under that same tuple. The first endpoint is sensitive to duplicate and merged regions; the second retains every qualified membership and is therefore merge-tolerant. Their ordering is a property of the thresholded bipartite graph. Neither endpoint estimates, lower-bounds, or upper-bounds the number of plants that a person could recover from a region; plant-count extraction was not measured.

Figure 1 separates inference from offline scoring and places the endpoints within the evidence chronology. Target truth can enter declared training or validation roles. Test masks and instance identifiers cannot enter proposal generation, ranking features, or queue allocation. Taxonomic identity, agricultural role, and management morphotype remain distinct axes. The experiments evaluate only the crop–weed role axis.

a Finite agricultural review queue



b Threshold-conditioned geometric endpoints



c Evidence chronology

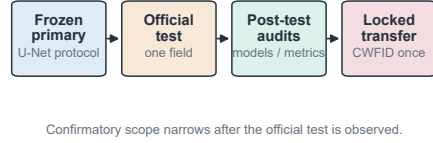


Figure 1: Evaluation contract and evidence chronology. (a) RGB inference produces a finite queue before test masks enter offline scoring. (b) At fixed K and matching thresholds, one-to-one support is the maximum candidate–instance matching cardinality and set coverage is the incident truth–instance union. These graph endpoints do not bound human recovery. (c) The frozen U-Net analysis preceded official-test access. Model, ranker, contract, and standard-metric audits followed that access. CWFID was separately locked and opened once.

2.2. Datasets, splits, and evidence classes

The source specialists used public SugarBeets2016 RGB images and masks [16]. Target development used 156 annotated 600×600 pixel WeedsGalore tiles from one maize field [17]. The official allocation contains 104 training, 26 validation, and 26 test tiles. Raster enumeration yielded 2,168 crop and 10,029 weed instances. The official test masks contained 256 crop and 1,712 weed instances.

The test split represents four acquisition dates (Table 1). The first three dates contribute eight tiles each. The last date contributes two. Date-block resampling treats the four dates as observed groups, but their physical independence is unknown. Capture indices cannot be mapped to verified field units. Image-bootstrap intervals therefore describe variation within this official image set, not a farm population.

Table 1: WeedsGalore official-test denominators by acquisition date.

Acquisition date	RGB tiles	Weed instances	Crop instances
25 May 2023	8	392	122
30 May 2023	8	440	55
6 June 2023	8	726	68
15 June 2023	2	154	11
Total	26	1,712	256

Evidence was assigned to three analysis classes before interpretation. The frozen primary experiment transferred five validation-selected U-Net checkpoints to CWFID. The matched-burden, strong-model, ranker-source, contract-grid, merge-aware, operator, and standard-metric analyses are post-test engineering audits on the official WeedsGalore split. Threshold sweeps of CWFID scores are post-lock diagnostics. Post-test and post-lock analyses are descriptive and exploratory. They cannot restore a previously unseen confirmatory status.

Cross-dataset evaluation used the official CWFID v1.0 release [18]. Its 60 RGB images and public annotations were fixed at source commit `176612f3`. The protocol fixed preprocessing, five existing U-Net checkpoints, validation-selected settings, $K = 20$, and endpoints before data access. All 242 upstream blobs were verified before one recorded opening. Weed truth units were eight-connected annotation components of at least 16 pixels. They are component proxies, not persistent plant identities. No CWFID label was used for fitting or selection.

2.3. Supervision ledger

Table 2 reports the supervision burden. Proposal generation is image-only during inference, but proposal operating-point commissioning uses target-training masks. Candidate role labels are also derived from dense masks. A 300-candidate ablation limits sampled training candidates while retaining all 2,436 labelled validation candidates. It is not a 300-label end-to-end regime.

2.4. Candidate formation, rankers, and model families

Two ResNet-18 U-Net-style source specialists were initialised with ImageNet-1K weights [19]. One estimated plant foreground and the other estimated crop and weed probabilities. Thresholding foreground probability

Table 2: Supervision and data roles. Test truth is restricted to offline scoring.

System or stage	Training supervision	Selection supervision	Test use
Source specialists	7,050 SugarBeets RGB masks	901 source masks for diagnostics	RGB inference only
Proposal commissioning	104 target dense masks	Training-mask objective and burden constraint	Offline scoring
Source-transfer rankers	9,887 target-training candidate labels	2,436 validation candidate labels	Queue formed be- fore scoring
U-Net and DeepLabV3+	104 target RGB dense masks	26 validation RGB dense masks	Queue formed be- fore scoring
Mask R-CNN	104 target RGB semantic and instance masks	26 validation semantic and instance masks	Queue formed be- fore scoring
CWFID transfer	No external labels	No external labels	60 annotations for locked scoring only
Two-operator review	No model training	Frozen 518-region queue and common instructions	Aggregate technical- review analysis

at T , applying eight-connectivity, and removing components below A pixels defined setting (T, A) . Full training details appear in the supplement.

The original operating point was $(0.95, 32)$. Proposal operating-point commissioning searched 12 settings on the 104 official-training tiles. Feasible settings produced no more than 100 candidates per tile on average. This limit bounded offline feature extraction and prevented unconstrained proposal expansion. It was deliberately wider than $K = 20$, which defined the downstream review capacity. Training role-qualified proposal recall was maximised, followed by lower burden, higher threshold, and larger area. The selected setting was $(0.70, 32)$.

Five candidate rankers used source weed score, geometry, role-plus-local features, frozen DINOv2 features, or component-masked DINOv2 features [20]. Logistic regularisation was selected on official validation. The burden-matched proposal audit tested four training schemes: commissioned-only, original-only, the union of both pools, and reciprocal cross-pool transfer. The reciprocal scheme trained on one pool and applied that ranker to the other pool.

Target semantic models comprised a ResNet-18 U-Net and DeepLabV3+ with a ResNet-50 encoder [21]. The instance model was Mask R-CNN with a

ResNet-50 FPN backbone [22]. U-Net used five training seeds. DeepLabV3+ and Mask R-CNN used three. Validation selected each checkpoint and component policy. All model families used the same $K = 20$ endpoint. Training used one NVIDIA H200 accelerator per seed.

2.5. Support, matching, and standard metrics

For truth instance I_j and candidate C_i , instance coverage is

$$\text{cov}(I_j, C_i) = \frac{|I_j \cap C_i|}{|I_j|}. \quad (1)$$

Spatial support requires at least one candidate with coverage of 0.50. Role-qualified support additionally requires 0.50 labelled-pixel coverage and 0.90 same-role purity. These values form the primary matching contract. A $3 \times 3 \times 3$ grid used instance coverage, labelled coverage, and purity values of 0.25, 0.50, and 0.75, with 0.90 replacing 0.75 at the highest purity level.

For a selected queue Q_K and primary threshold tuple $\tau = (0.50, 0.50, 0.90)$, let E_τ contain every candidate–weed-instance pair meeting the instance-coverage, labelled-coverage, and same-role-purity thresholds. Set coverage is $|\{I_j : \exists C_i \in Q_K, (C_i, I_j) \in E_\tau\}|$. One-to-one support is the maximum-cardinality matching on the same thresholded graph. Candidate and instance identifiers resolve ties. The difference between these numerators is the merge-capacity gap after duplicate memberships are removed. Candidate precision is the fraction of selected regions satisfying the role-qualified weed contract. Every reported endpoint is indexed by K and τ ; changing either changes the estimand.

Crop exposure is the fraction of tiles with a queued region overlapping crop truth. We report any positive overlap and candidate-area overlap thresholds of 1% and 10%. Standard diagnostics include mAP over IoU 0.50–0.95, AP50, AP75, AR100, and queue AR20. A PQ-style diagnostic is computed on the predicted-weed subset of the $K = 20$ queue. It is not full panoptic quality because background and crop segments do not form a complete panoptic partition.

2.6. Matched burden and uncertainty

Commissioning changes candidate support and burden. A common per-tile capacity used

$$k_i = \min(20, n_i^{\text{original}}, n_i^{\text{commissioned}}) \quad (2)$$

for test tile i . Both settings contributed 471 candidates. Ranker-free proposal-set ceilings and per-tile $K = 20$ one-to-one oracles separate formation capacity from achieved ranking.

The primary model comparison uses 20,000 crossed bootstrap replicates. Each replicate samples a training seed within each model and resamples the 26 test images with replacement. The two prespecified one-to-one contrasts are DeepLabV3+ minus U-Net and Mask R-CNN minus U-Net. We report 95% intervals and Bonferroni-adjusted familywise 97.5% intervals. A date-block plus seed bootstrap is a sensitivity analysis over four observed dates. No interval is interpreted as a farm-population interval.

2.7. Two-operator technical review

Two different human operators independently reviewed the frozen 518-region U-Net queue. They used common written instructions and 12 practice items from validation data. They did not discuss formal items, view peer responses, use automated labels, or conduct external searches. Acquisition date, tile identity, candidate identifiers, scores, ranks, model identity, and mask-derived truth were hidden.

The operators recorded primary content, weed reviewability, crop presence, localisation usability, and five-point confidence. Timing was not collected or analysed. Exact agreement, Cohen’s κ , Gwet’s AC1, and scene-cluster intervals were computed. Linear and quadratic weighted κ were added for ordered fields. Human consensus and mask qualification were compared in a three-by-two contingency table over all 518 regions. Binary precision, recall, F1, Jaccard, and accuracy were computed only on regions with determinate consensus (both “yes” or both “no”); disputed or uncertain regions were retained in the contingency table and excluded from those binary denominators.

3. Results

3.1. Commissioning expanded raw support, while queued effects depended on ranker training

The original setting produced 802 test candidates, or 30.85 per tile. The commissioned setting produced 2,351, or 90.42 per tile. Spatial proposal recall increased from 0.2973 to 0.5713. Role-qualified proposal recall increased from 0.1928 to 0.3026. These ranker-free changes establish greater raw geometric support.

Table 3: Commissioned-minus-original effects at the common 471-candidate burden under the primary matching contract.

Ranker training pool	Precision	Spatial	Set coverage	One-to-one
Commissioned only	−0.0679	+0.0818	+0.0374	+0.0187
Original only	−0.3503	−0.1268	−0.0695	−0.0514
Union	−0.0701	+0.0847	+0.0380	+0.0210
Reciprocal cross-pool	−0.3482	−0.1460	−0.0847	−0.0625

At the common 471-candidate burden, the commissioned-only ranker increased spatial recall by 0.0818, set coverage by 0.0374, and one-to-one recall by 0.0187. Precision decreased by 0.0679. The primary-contract one-to-one change was 0.0210 with the union ranker, −0.0514 with the original-only ranker, and −0.0625 under reciprocal transfer (Table 3). Commissioning therefore did not produce a ranker-independent queue improvement.

The same dependence appeared across 27 contracts (Figure 2). Spatial support increased in all contracts for commissioned-only and union rankers. One-to-one support increased in 15 and 16 contracts, respectively. It decreased in every contract for original-only and reciprocal rankers. Under the primary contract, the original and commissioned proposal-set ceilings were 0.1928 and 0.3026. Their per-tile $K = 20$ oracles were 0.1396 and 0.1618. Achieved one-to-one support was 0.0970 and 0.1157.

3.2. Set coverage exposed a merge-capacity gap

The 518-region U-Net queue contained 170 regions with no qualified weed instance, 286 with one, and 62 with two or more. One merged region covered nine qualified instances. The union of qualified memberships contained 453 distinct instances. One-to-one matching retained 348. Set coverage was therefore 0.2646, whereas one-to-one support was 0.2033. The 105-instance difference arose from merged capacity; no duplicate membership remained after the set union.

Among 301 consensus-reviewable regions, set coverage was 0.2021 and one-to-one support was 0.1454 under the primary threshold tuple. Their 97-instance graph gap again reflected merges. Reviewability increased with geometric multiplicity: 0.3059 for no qualified instance, 0.6818 for one, 0.8140 for two, and 1.0000 for three or more. Merged regions can be useful evidence, but this association does not convert set coverage into a human-recovery

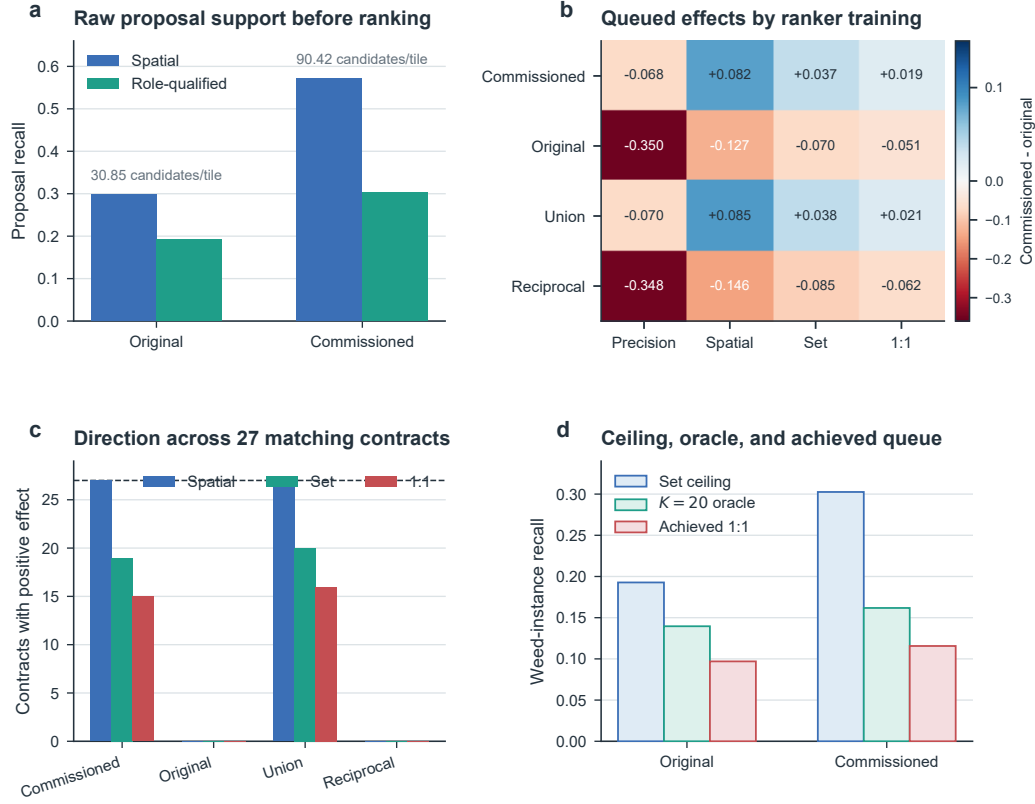


Figure 2: Proposal and ranker sensitivity. (a) Raw spatial and role-qualified proposal support before ranking. (b) Commissioned-minus-original effects under four ranker-training schemes. (c) Number of positive effects across 27 matching contracts. (d) Ranker-free set ceiling, per-tile $K = 20$ oracle, and achieved one-to-one support for the commissioned-only ranker.

endpoint because the operators did not count independently recoverable plants.

3.3. Standard metrics and crossed uncertainty changed the model comparison

Mask R-CNN had the highest one-to-one point estimate at $K = 20$: 0.2159 ± 0.0038 across three seeds. U-Net reached 0.2027 ± 0.0057 across five seeds, and DeepLabV3+ reached 0.1924 ± 0.0032 across three. The crossed seed-by-image contrast was $+0.0132$ for Mask R-CNN versus U-Net. Its 95% interval was -0.0009 to 0.0300 . The familywise interval was -0.0027 to 0.0329 . DeepLabV3+ minus U-Net was -0.0103 , with a 95% interval of -0.0237 to 0.0026 . Neither contrast excluded zero.

Table 4: WeedsGalore instance metrics on the same predicted-weed regions. Each cell reports the seed mean with the sample SD in parentheses.

Model	Seeds	One-to-one $K = 20$	mAP	AP50	AR100@50	Queue AR20@50
U-Net	5	0.2027 (0.0057)	0.0853 (0.0037)	0.2616 (0.0157)	0.3412 (0.0207)	0.1672 (0.0070)
DeepLabV3+	3	0.1924 (0.0032)	0.0499 (0.0026)	0.1937 (0.0060)	0.2950 (0.0081)	0.1456 (0.0045)
Mask R-CNN	3	0.2159 (0.0038)	0.1064 (0.0082)	0.3189 (0.0248)	0.3904 (0.0266)	0.1861 (0.0079)

The date-block plus seed interval was 0.0023 to 0.0341 for Mask R-CNN minus U-Net and -0.0244 to 0.0133 for DeepLabV3+ minus U-Net. These four-date intervals were more favourable to Mask R-CNN but remain conditional on the observed dates and limited seed counts. We therefore report Mask R-CNN as the highest point estimate, not as a statistically established winner.

Standard metrics on the same predictions also favoured Mask R-CNN (Table 4). Its mAP was 0.1064 and AP50 was 0.3189. U-Net reached 0.0853 and 0.2616, respectively. At the queue limit, Mask R-CNN reached AR20@50 of 0.1861 and a PQ-style score of 0.1936. The matching-contract grid was less uniform: Mask R-CNN ranked first in 21 of 27 contracts, while U-Net ranked first in six. U-Net led when instance coverage was 0.75 with labelled coverage of 0.25 or 0.50.

Failure cases show why the endpoints diverge (Figure 4). A split plant can occupy several reviews. A merged candidate can support several truth instances but create one record. Background regions and crop–weed mixtures consume capacity without satisfying the primary role contract.

3.4. Operators agreed on categories, but human and geometric criteria differed

Both operators completed all 518 regions. Weed-reviewability agreement was 0.9421, Cohen’s κ was 0.8930, and Gwet’s AC1 was 0.9206. The scene-bootstrap interval for AC1 was 0.8918 to 0.9483. Primary content, crop presence, and localisation AC1 values were 0.9355, 0.9341, and 0.9243.

Confidence matched exactly for 0.9440 of items. Linear weighted κ was 0.9366 and quadratic weighted κ was 0.9589. Mean confidence was 3.523 for one operator and 3.510 for the other. Both medians were 4, with interquartile ranges of 3–4. Among 445 determinate localisation pairs, exact agreement was 0.9955 and quadratic weighted κ was 0.9966.

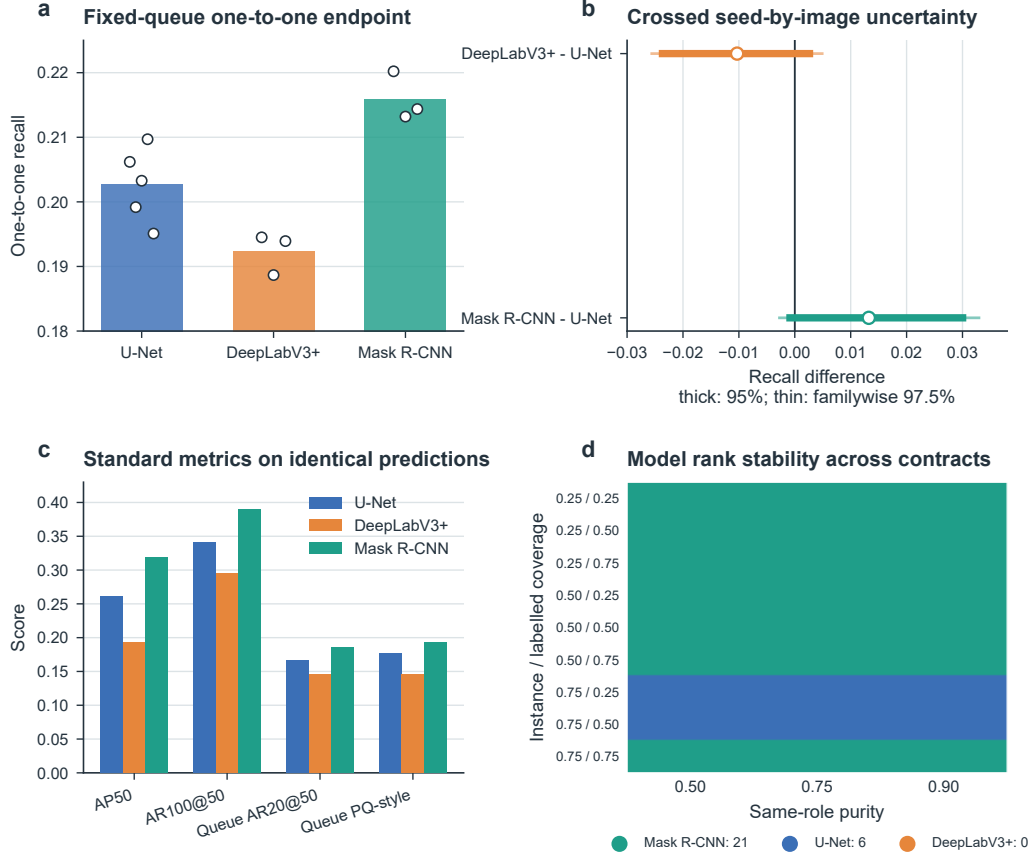
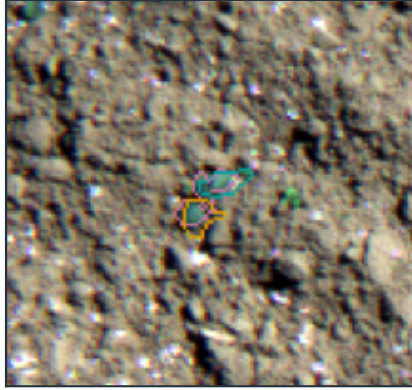


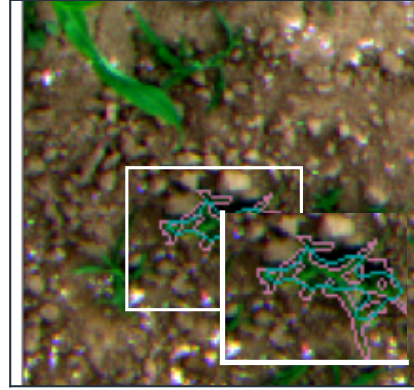
Figure 3: Model comparison on the 26 WeedsGalore test tiles. (a) Seed-level one-to-one support. (b) Crossed seed-by-image contrasts with U-Net. Thick lines show 95% intervals; thin lines show Bonferroni-adjusted familywise 97.5% intervals. (c) Standard metrics on identical predictions. (d) Best one-to-one model across 27 matching contracts.

The operators reached consensus that 301 regions were reviewable and 143 were not. Seventy-four were disputed or contained uncertainty. Mask qualification agreed with consensus reviewability on 249 positive and 76 negative regions. It produced 67 mask-positive/human-negative and 52 mask-negative/human-positive regions. Thirty-nine unresolved regions were mask-positive and 35 were mask-negative. On the 444 determinate-consensus regions (85.71% of all items), mask precision was 0.7880, recall was 0.8272, F1 was 0.8071, Jaccard was 0.6766, and accuracy was 0.7320. Human reviewability and fixed geometry are related but non-equivalent evidence.

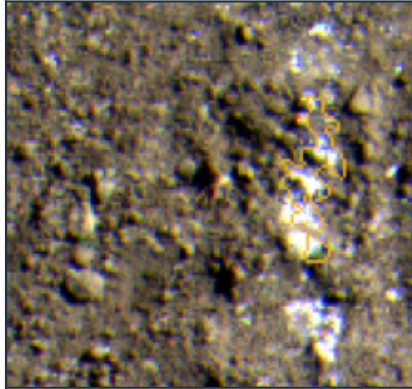
a Split: one weed, two fragments
30 May, image 0649



b Merge: one region, five weeds
6 June, image 0176



c Background component
25 May, image 0109



d Crop-weed mixture
6 June, image 0770



Cyan: candidate boundary | Magenta: weed truth | Green: crop truth | Yellow: background or mixed region

Figure 4: Representative proposal failures from the official WeedsGalore test split. (a) A weed split into two candidates. (b) One candidate covering five weed instances, with a local enlargement. (c) A background component. (d) A crop–weed mixture. Truth contours are used only for offline illustration.

3.5. Locked CWFID transfer failed before queue formation

The locked primary CWFID evaluation contained 60 images and 331 weed-component proxies. Every U-Net checkpoint produced zero candidates at its frozen validation-selected threshold. Thus no queue item or component

recovery was observed. The empirical all-zero bootstrap was degenerate and is not reported as a 0–0 population interval. For candidate occurrence across 60 images, the one-sided 95% binomial reference upper bound was 0.0487. This bound is not a component-recall or farm-population guarantee.

Post-lock diagnostics retained the frozen result and did not select an external threshold. Image-macro plant-pixel AUROC ranged from 0.3871 to 0.6374 across checkpoints. Pixel AP ranged from 0.0438 to 0.0715. These scores show weak and checkpoint-variable ranking. Calibration bins placed nearly all pixels below 0.05, while observed plant prevalence in the lowest bin was about 0.055.

At diagnostic threshold 0.01, spatial recall ranged from 0.0363 to 1.0000. Role-qualified one-to-one support was nonzero for only one checkpoint and reached 0.0423. Some thresholds generated broad or mixed regions, including one checkpoint with spatial recall 1.0 but qualified recall 0. This pattern does not isolate a single failure mechanism. Threshold mismatch, role semantics, scale, viewpoint, crop type, and annotation proxies may all contribute.

4. Discussion

The evaluation contract reveals information that candidate discrimination and standard instance scores do not provide alone. AP, AR, and PQ-style diagnostics measure localisation, segmentation, ranking, and duplicate errors under established conventions. The finite-queue contract adds an explicit agricultural capacity and decomposes losses before and after that capacity. Set coverage and one-to-one support further separate merged capacity from duplicate capacity.

The merge-aware analysis distinguishes two graph quantities under the primary threshold tuple and $K = 20$. One-to-one support is the maximum number of candidate–plant pairs that can be assigned without reusing either node. Set coverage is the truth-instance union over all qualified edges and therefore retains multiple memberships of a merged candidate. Their 105-instance difference quantifies merge capacity in this thresholded graph. It is not a range for human performance: geometric overlap does not establish visual separability, and the operator protocol did not collect plant counts.

Proposal operating-point commissioning improved ranker-free spatial and role-qualified support. Its queued benefit depended on which proposal distribution trained the ranker. Commissioned-only and union training produced positive median effects, whereas original-only and reciprocal transfer reversed

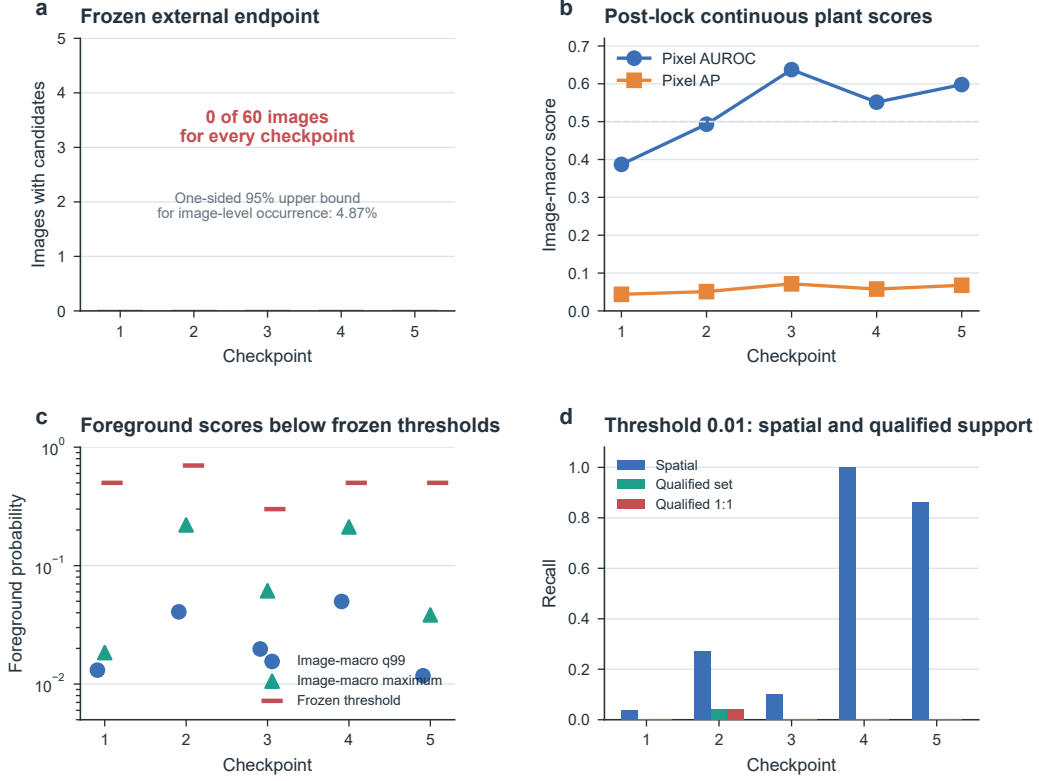


Figure 5: Locked CWFID transfer and post-lock diagnostics. (a) Frozen checkpoints produced no candidates in any image; the one-sided reference bound concerns image-level candidate occurrence only. (b) Continuous plant-pixel AUROC and AP varied across checkpoints. (c) foreground scores remained below frozen validation thresholds. (d) At diagnostic threshold 0.01, spatial coverage could be high while role-qualified support remained absent or small.

them. This result distinguishes a proposal ceiling from an achieved queue. A larger candidate pool can contain more threshold-qualified support while remaining poorly ordered by a mismatched ranker.

The model comparison is similarly conditional. Mask R-CNN led most matching contracts and all reported standard metrics. Crossed uncertainty still included zero for its one-to-one contrast with U-Net. Only three Mask R-CNN seeds and five U-Net seeds were available. The four acquisition dates came from one field. These results support a model-ranking hypothesis for future evaluation, not a general population claim.

The two-operator study establishes that the review categories can be

applied consistently by these two operators. It does not establish operator-population performance, plant-count recovery, review speed, or labour economics. Disputed and uncertain regions would enter the escalation branch of the workflow. On determinate consensus items, the human-mask Jaccard index of 0.6766 also shows why geometric qualification should remain an offline measurement rather than a surrogate for human usefulness.

The CWFID experiment defines a clear transfer boundary. The frozen system failed before queue construction. Continuous diagnostics found some ranking signal in selected checkpoints but weak AP and unstable threshold behaviour. The evidence cannot assign the failure solely to calibration. External evaluation covered U-Net because those five checkpoints and their thresholds were fixed before CWFID opening. Mask R-CNN and semantic reordering were developed after official-test access and were not inserted retrospectively into the locked protocol.

The contract can enter an executable annotation-QA workflow. A team first fixes a candidate source, role criterion, review budget, and escalation rule. It then measures proposal ceilings, achieved set and one-to-one support, crop exposure, and standard AP/AR. Ranker-source and contract grids identify fragile improvements. A separately locked acquisition tests whether the queue still forms. Timing, plant-count extraction, and cost can be added as independent workflow endpoints when collected.

4.1. Limitations

WeedsGalore supplies one field and four acquisition dates. Capture groups do not identify independent plots. All post-test WeedsGalore extensions are exploratory engineering audits. Image and date-block intervals do not support farm-level inference. CWFID differs in crop, sensor geometry, scale, and annotation contract, and its components are plant proxies. The operator study involved two fixed individuals and omitted timing and recoverable-plant counts. The $K = 20$ capacity is experimental rather than economically optimised. Public weights and datasets support computational replay, but full retraining requires the upstream data and model assets listed in the archive.

5. Conclusion

Bounded instance-support evaluation connects crop-weed predictions to a finite offline review queue. Spatial and role-qualified proposal support locate

formation losses. At fixed queue capacity and matching thresholds, set coverage and one-to-one matching expose merge capacity under alternative graph assignments. Contract and ranker grids reveal conditional improvements, while crossed uncertainty prevents seed variation from being hidden. Standard instance metrics remain complementary. The locked external test shows that an internally stable model may fail before producing any review item. Together, these components form an auditable quality-assurance protocol for comparing agricultural vision queues without assigning them unmeasured deployment value.

Data availability

SugarBeets2016, WeedsGalore, and CWFID are public datasets available from their cited sources. The reproducibility release records source URLs, access dates, release identifiers, licence information, versions, and integrity hashes. Public imagery remains with its original repositories. Derived aggregate tables, matching outputs, and verification manifests accompany the code release.

Code availability

Versioned reproducibility materials are maintained at <https://github.com/Fytap/agrispec-vlm-reproducibility>. The manuscript-linked v1.3.1 evidence layer is fixed at the immutable commit https://github.com/Fytap/agrispec-vlm-reproducibility/tree/6930f7b48995054095dbaaeebef4f155029c107a/reproducibility/v1_3_1. The compact verifier replays archived tables, figures, and numerical invariants. It does not retrain every model from raw imagery. Full end-to-end reproduction uses the supplied training and evaluation scripts, public data instructions, environment specification, checkpoint hashes, and model acquisition records. Failed runs and their causes are preserved.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used AI-assisted tools for code review and language review, including checks of sentence clarity, grammar, and wording. The authors reviewed and edited all affected content and take full responsibility for the content of the publication.

Ethics statement. No external participants were recruited and no personal data were collected; human-participant ethics approval and informed consent were not applicable.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- [1] D. C. Slaughter, D. K. Giles, D. Downey, Autonomous robotic weed control systems: A review, *Computers and Electronics in Agriculture* 61 (1) (2008) 63–78. doi:10.1016/j.compag.2007.05.008.
- [2] S. M. Swinton, Economics of site-specific weed management, *Weed Science* 53 (2) (2005) 259–263. doi:10.1614/WS-04-035R2.
- [3] J. J. Casanova, N. T. Bergmann, J. E. R. Kalin, G. C. Heineck, I. C. Burke, A comparison of protocols for high-throughput weeds mapping, *Smart Agricultural Technology* 12 (2025) 101076. doi:10.1016/j.atech.2025.101076.
- [4] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C. L. Zitnick, Microsoft COCO: Common objects in context, in: *Computer Vision – ECCV 2014*, 2014, pp. 740–755. doi:10.1007/978-3-319-10602-1_48.
- [5] A. Kirillov, K. He, R. Girshick, C. Rother, P. Dollár, Panoptic segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9404–9413. doi:10.1109/CVPR.2019.00963.
- [6] R. Jena, L. Zhorniyak, N. Doiphode, P. Chaudhari, V. Buch, J. Gee, J. Shi, Beyond mAP: Towards better evaluation of instance segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 11309–11318.
- [7] L. Chen, Y. Wu, J. Stegmaier, D. Merhof, SortedAP: Rethinking evaluation metrics for instance segmentation, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2023, pp. 3923–3929.
- [8] J. Hosang, R. Benenson, P. Dollár, B. Schiele, What makes for effective detection proposals?, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 38 (4) (2016) 814–830. doi:10.1109/TPAMI.2015.2465908.

- [9] P. Kar, H. Narasimhan, P. Jain, Surrogate functions for maximizing precision at the top, in: Proceedings of the 32nd International Conference on Machine Learning, Vol. 37 of Proceedings of Machine Learning Research, PMLR, 2015, pp. 189–198.
URL <https://proceedings.mlr.press/v37/kar15.html>
- [10] R. Sheikh, A. Milioto, P. Lottes, C. Stachniss, M. Bennewitz, T. Schultz, Gradient and log-based active learning for semantic segmentation of crop and weed for agricultural robots, in: 2020 IEEE International Conference on Robotics and Automation, 2020, pp. 1350–1356. doi:10.1109/ICRA40945.2020.9196722.
- [11] B. M. van Marrewijk, C. Dandjinou, D. J. A. Rustia, N. F. Gonzalez, B. Diallo, J. Dias, P. Melki, P. M. Blok, Active learning for efficient annotation in precision agriculture: a use-case on crop–weed semantic segmentation, arXiv preprint arXiv:2404.02580 (2024). doi:10.48550/arXiv.2404.02580.
- [12] Y. Geifman, R. El-Yaniv, Selective classification for deep neural networks, in: Advances in Neural Information Processing Systems, Vol. 30, 2017, pp. 4878–4887.
URL <https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00b67e46540e2b0cf1-Abstract.html>
- [13] Y. Geifman, R. El-Yaniv, Selectivenet: A deep neural network with an integrated reject option, in: Proceedings of the 36th International Conference on Machine Learning, Vol. 97 of Proceedings of Machine Learning Research, 2019, pp. 2151–2159.
URL <https://proceedings.mlr.press/v97/geifman19a.html>
- [14] H. Mozannar, D. Sontag, Consistent estimators for learning to defer to an expert, in: Proceedings of the 37th International Conference on Machine Learning, Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 7076–7087.
URL <https://proceedings.mlr.press/v119/mozannar20b.html>
- [15] H. Mozannar, H. Lang, D. Wei, P. Sattigeri, S. Das, D. Sontag, Who should predict? exact algorithms for learning to defer to humans, in: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, Vol. 206 of Proceedings of Machine Learning Research, PMLR, 2023, pp. 10520–10545.
URL <https://proceedings.mlr.press/v206/mozannar23a.html>
- [16] N. Chebroly, P. Lottes, A. Schaefer, W. Winterhalter, W. Burgard, C. Stachniss, Agricultural robot dataset for plant classification, localization and mapping on sugar beet fields, The International Journal of Robotics Research 36 (10) (2017) 1045–1052. doi:10.1177/0278364917720510.
- [17] E. Celikkan, T. Kunzmann, Y. Yeskaliyev, S. Itzerott, N. Klein, M. Herold, Weeds-galore: A multispectral and multitemporal uav-based dataset for crop and weed segmentation in agricultural maize fields, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2025, pp. 4767–4777.
URL https://openaccess.thecvf.com/content/WACV2025/html/Celikkan_WeedsGalore_A_Multispectral_and_Multitemporal_UAV-Based_Dataset_for_Crop_and_WACV_2025_paper.html
- [18] S. Haug, J. Ostermann, A crop/weed field image dataset for the evaluation of computer vision based precision agriculture tasks, in: Computer Vision – ECCV 2014 Workshops, 2015, pp. 105–116. doi:10.1007/978-3-319-16220-1_8.
- [19] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition,

- 2016, pp. 770–778. doi:10.1109/CVPR.2016.90.
- [20] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, R. Howes, P.-Y. Huang, H. Xu, V. Sharma, S.-W. Li, W. Galuba, M. Rabbat, M. Assran, N. Ballas, G. Synnaeve, I. Misra, H. Jégou, J. Mairal, P. Labatut, A. Joulin, P. Bojanowski, Dinov2: Learning robust visual features without supervision, Transactions on Machine Learning Research (2024).
URL <https://mlanthology.org/tmlr/2024/oquab2024tmlr-dinov2/>
 - [21] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder–decoder with atrous separable convolution for semantic image segmentation, in: Proceedings of the European Conference on Computer Vision, 2018, pp. 801–818.
URL https://www.ecva.net/papers/eccv_2018/papers_ECCV/html/Liang-Chieh_Chen_Encoder-Decoder_with_Atrous_ECCV_2018_paper.php
 - [22] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask R-CNN, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2961–2969. doi:10.1109/ICCV.2017.322.
URL https://openaccess.thecvf.com/content_iccv_2017/html/He_Mask_R-CNN_ICCV_2017_paper.html